

Supplementary Material—360° RAYS

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OVERVIEW

This supplementary document provides additional material for 360° RAYS. The document presents extended qualitative results across multiple datasets, showing how different ordering and layout choices affect interpretation and how, by using RAYS, we get various analyses.

1 RAYS VISUALISATIONS

The following sections contain 360° RAYS visualisations of different datasets, together with possible analyses derived from them.

1.1 Pima Indians Diabetes Dataset [3]

Dataset details: 9 features, 2 classes, 768 points.

Number of empty sectors in last ring (9-dimensional subspace): 210

1.1.1 Ascending Order of CoV

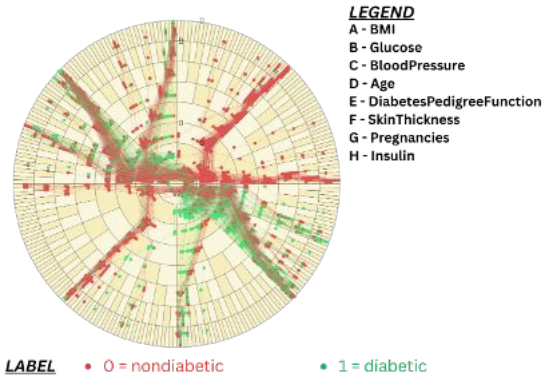


Figure 1: Pima Indians Diabetes Dataset in ascending order of CoV (using female patient records only).

Figure 1 shows the Pima Indians Diabetes Dataset using ascending order of CoV. This ordering places the lower-variance dimensions closer to the centre, giving a stable overview of how the points branch across the subspaces before reaching the outer rings. The plot supports comparison of the two classes by showing where their paths overlap and where they begin to separate across the occupied orthants.

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1.1.2 Descending Order of CoV

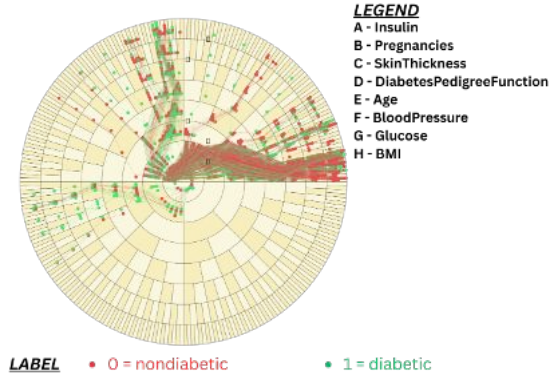


Figure 2: Pima Indians Diabetes Dataset in descending order of CoV (using female patient records only).

Figure 2 shows the same dataset in descending order of CoV, moving the most variable dimensions toward the centre of the visualisation. Compared to the ascending-order view, the major splits appear earlier in the hierarchy, making it easier to inspect if the highly variable features create clearer class separation.

1.2 Blood Transfusion Service Center Dataset [5]

Dataset details: 4 features, 2 classes, 748 points.

Number of empty sectors in last ring (4-dimensional subspace): 11

1.2.1 Ascending Order of CoV

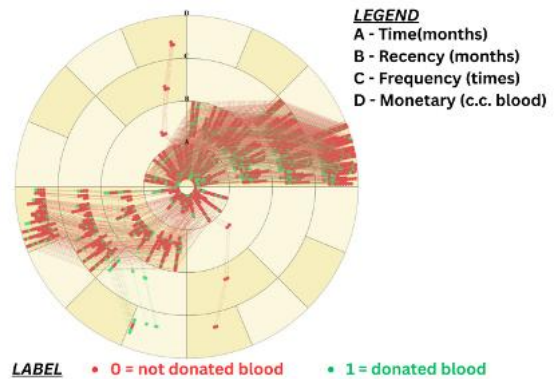


Figure 3: Blood Transfusion Service Center Dataset subspaces in ascending order of CoV.

Figure 3 shows the Blood Transfusion Service Center Dataset using ascending order of CoV of dimensions. At first glance, we observe that the number of points belonging to Class 0 is more than that of Class 1. For all the points, the value of the coordinate of Dimension

2, i.e., Recency, is comparatively. There does not seem to be any differentiating feature for both classes in the dataset.

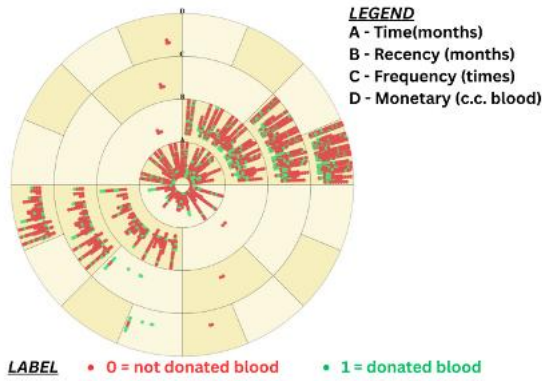


Figure 4: Blood Transfusion Service Center Dataset subspaces in ascending order of CoV, with edge connections removed.

Figure 4 shows a modified version of the same ascending-order layout in which the connection edges are removed to analyse sector-wise point details more clearly. This makes the distribution of points in each sector easier to observe. And as observed in figure 3, there is no differentiating feature or value for both the classes, as multiple points belonging to different classes have the same value (points on the same radial direction in each sector).

1.2.2 Descending Order of CoV

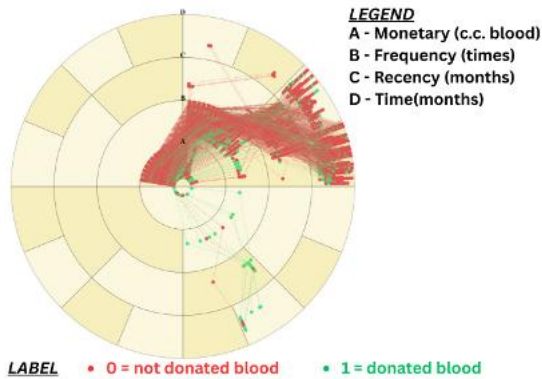


Figure 5: Blood Transfusion Service Center Dataset subspaces in descending order of CoV.

Figure 5 shows the Blood Transfusion Service Center Dataset using descending order of CoV of dimensions. Compared to the ascending-order view, the branching structure appears different, and the orthants occupied in this setting are different compared to the ascending order of CoV for dimension ordering. However, in this case, most of the points are moving towards the same direction despite putting the high variant dimension in the innermost ring, suggesting that the data has a strong directional tendency along a particular dimension (innermost dimension). This also suggests that the innermost feature is skewed or shifted, but not evenly distributed around zero.

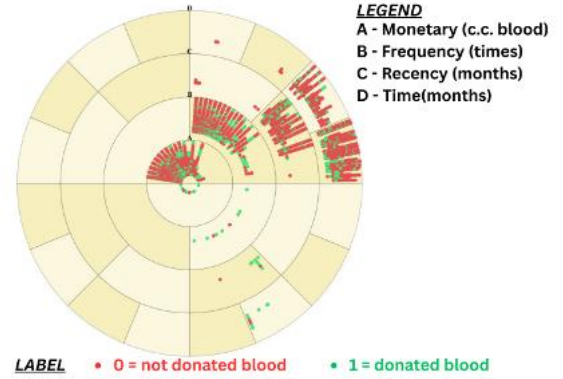


Figure 6: Blood Transfusion Service Center Dataset subspaces in descending order of CoV, with edge connections removed.

Figure 6 shows the same descending-order layout with the edge connections removed so that the occupancy and class distribution of individual sectors can be analysed more clearly. By observing the occupied sector, we see that the data is indeed favouring one extreme value, suggesting that the data is shifted.

1.3 Breast Cancer Dataset [4]

Dataset details: 9 features, 2 classes, 699 points.

Number of empty sectors in last ring (9-dimensional subspace): 376

1.3.1 Ascending Order of CoV

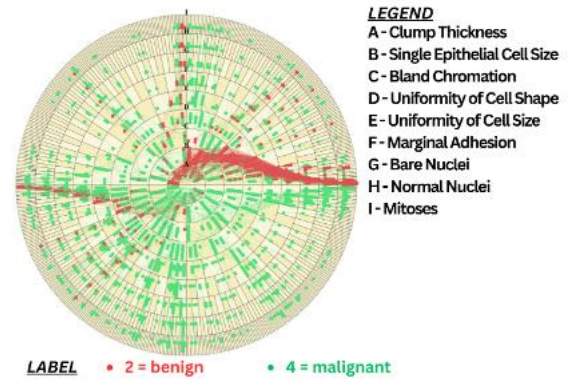


Figure 7: Breast Cancer Dataset subspaces in ascending order of CoV.

Figure 7 shows the Breast Cancer Dataset in ascending order of CoV, with the full subspace structure and empty sectors retained so that sparsity remains visible across the hierarchy. We observe and evenly distribution of the points across the space; however, in particular class points, especially for class 2, the points are concentrated towards particular orthants only.

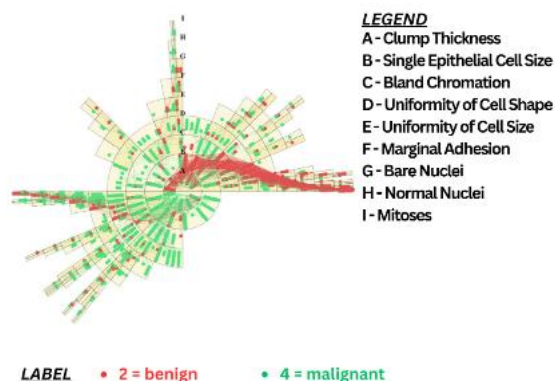


Figure 8: Compact view of the Breast Cancer Dataset subspaces in ascending order of CoV.

Figure 8 provides a compact view of the same ascending-order arrangement. This shows that certain features are critical and shows early pruning of classes. For example, Ring B shows pruning for the points belonging to Class 4.

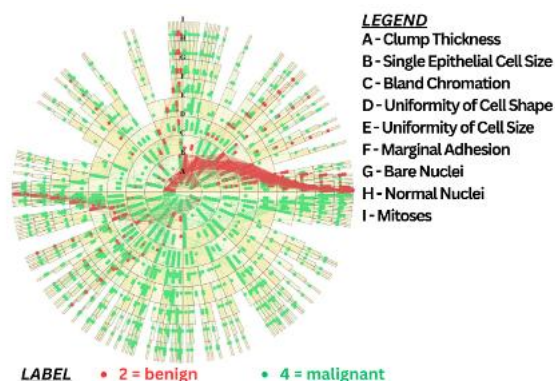


Figure 9: Breast Cancer Dataset with collapsed empty sectors in ascending order of CoV.

Figure 9 removes empty sectors to emphasise the occupied paths and reveal the denser regions more directly.

1.3.2 Descending Order of CoV

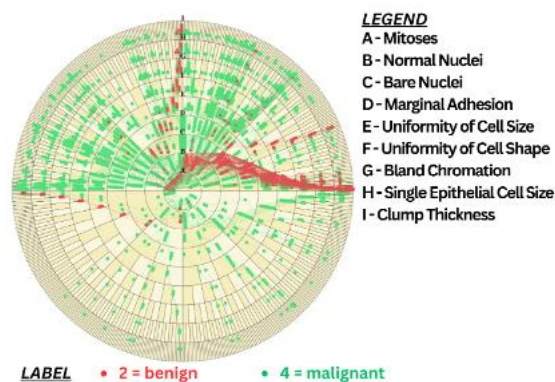


Figure 10: Breast Cancer Dataset subspaces in descending order of CoV.

Figure 10 shows the Breast Cancer Dataset in descending order of CoV. With higher-variation dimensions placed closer to the centre,

the early branching becomes more visually dominant and separates major groups sooner. However, we see a greater concentration of points in the upper half, presenting the skewness of the data in the innermost ring (dimension).

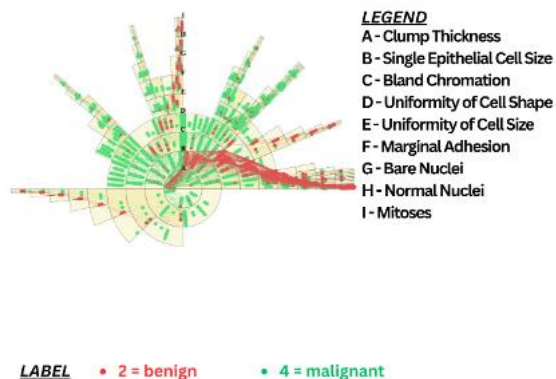


Figure 11: Compact view of the Breast Cancer Dataset subspaces in descending order of CoV.

Figure 11 represents the same descending-order structure in a compact view, showing the pruning of features in an intuitive manner.

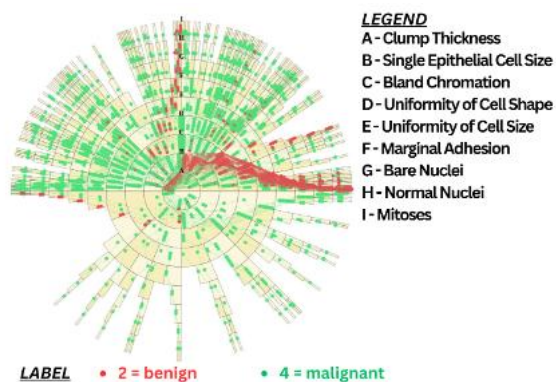


Figure 12: Breast Cancer Dataset with collapsed empty sectors in descending order of CoV.

Figure 12 highlights only non-empty sectors to emphasise the occupied paths and reveal the denser regions more directly.

1.4 Vertebral Column Dataset [1]

Dataset details: 6 features, 2 classes, 310 points.

Number of empty sectors in last ring (6-dimensional subspace): 44

1.4.1 Ascending Order of CoV

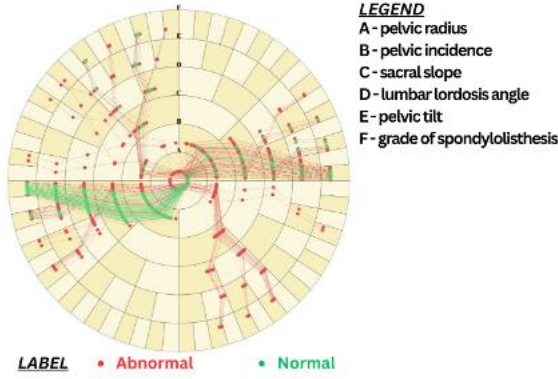


Figure 13: Vertebral Column Dataset subspaces in ascending order of CoV.

Figure 13 shows the ascending-order hierarchy, providing a clear view of the distribution of points and classes.

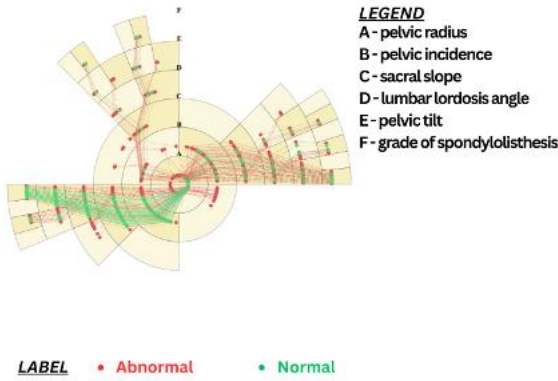


Figure 14: Compact view of the Vertebral Column Dataset subspaces in ascending order of CoV.

Figure 14 shows the Vertebral Column Dataset in ascending order of CoV using a compact layout. We see that the majority of the sectors are mixed/impure sectors; it is hard to classify the points based on certain values of the coordinates (the orthants they belong to).

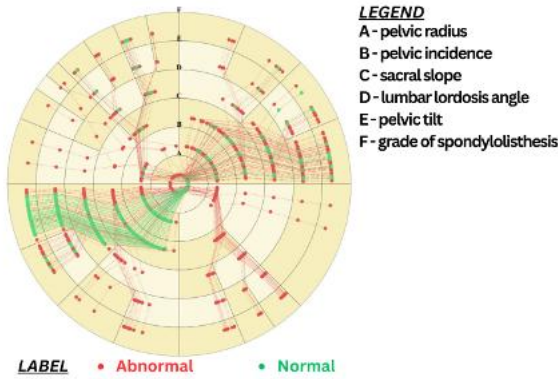


Figure 15: Vertebral Column Dataset with collapsed empty sectors and refactored angles in ascending order of CoV.

Figure 15 removes empty sectors and refactors the angles without breaking the hierarchical structure of the rings and the sectors to provide more area to the points lying in outer rings, giving an enhanced view of the distribution of the points in each sector, regardless of the dimension. We observe that the branching and the direction remain the same, keeping the initial analysis intact and valid.

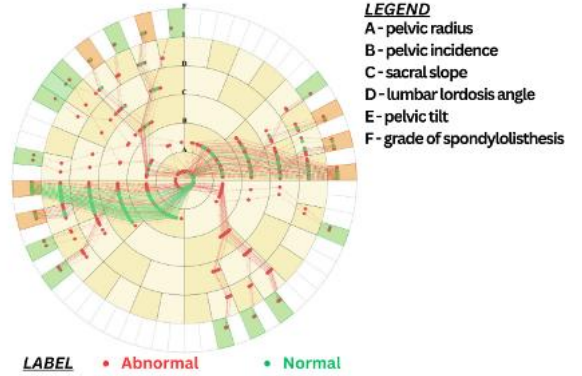


Figure 16: Vertebral Column Dataset subspaces in ascending order of CoV, with sector purity in last ring.

Figure 16 adds a class purity label to support detailed inspection of sector/orthant identities and their corresponding class distributions in the outermost ring, making it easier to understand the number of pure and impure orthants in R^d space, at a glance, without having to track the complete path of the points from innermost to outermost ring. Green colored sectors represent pure sectors, orange represent impure/mixed sectors and white are empty sectors.

1.4.2 Descending Order of CoV

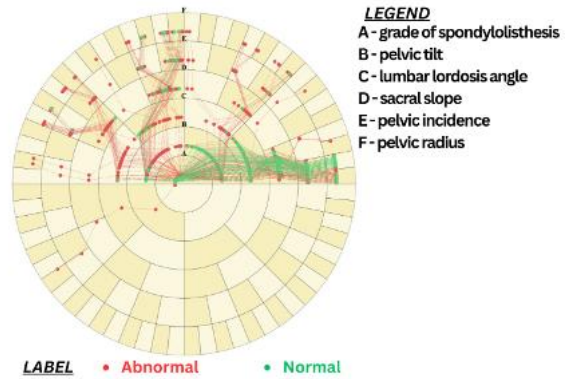


Figure 17: Vertebral Column Dataset subspaces in descending order of CoV.

Figure 17 shows the descending-order hierarchy showing the distribution of data points in the feature with a high coefficient of variation. It depicts that the points lie at an extremity, suggesting a skewness or shift in the data. Unlike the ascending order of CoV for ring ordering, the data is unevenly spread and is clustered towards one half of the space.

Along with this, we also observe, as compared to Figure 13, Figure 17 which follows a descending order of CoV for subspace ordering, shows that the number of orthants used in both the arrangements are visually different for the intermediate rings. We can therefore find the change in orthant usage. A greater change

in the number of orthants in different orderings of the features suggests that the feature is context-sensitive as compared to a smaller change, which means that feature contribution is stable across orders, and the feature can act as an independent separator (property t_7). Visually, by looking at the percentage of orthant utilization, we can interpret the importance of a feature in the dataset. However, this is limited to data distribution, which might not necessarily give a meaningful interpretation of the split, but this data distribution can also be seen by doing sector analysis (property t_5).

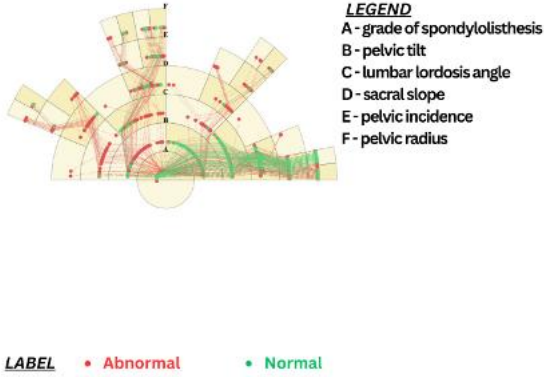


Figure 18: Compact view of the Vertebral Column Dataset subspaces in descending order of CoV.

Figure 18 shows the Vertebral Column Dataset in descending order of CoV using a compact layout. This ordering moves the most variable dimensions inward, making the main splits easier to recognise near the centre of the plot. The compact view enhances the analysis by quickly showing the class purity-based pruning at an early stage.

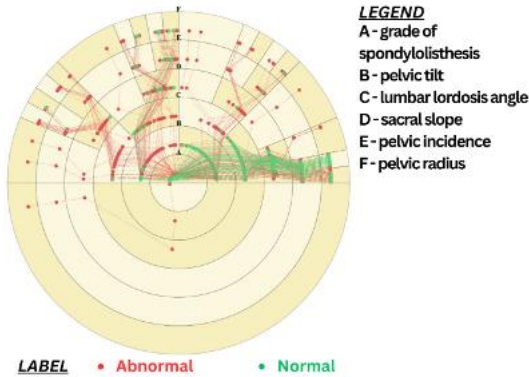


Figure 19: Vertebral Column Dataset with collapsed empty sectors and refactored angles in descending order of CoV.

Figure 19 simplifies the view by removing empty sectors and refactoring the area allocated to each sector, similar to Figure 15.

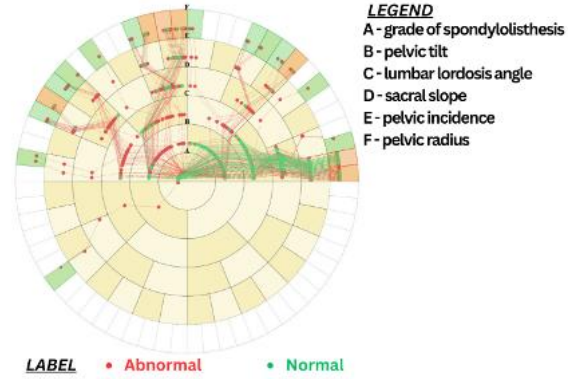


Figure 20: Vertebral Column Dataset subspaces in descending order of CoV, with class purity highlighted in the last ring.

Figure 20 further highlights class purity in the outer ring, making it easier to identify subspaces that are predominantly associated with a single class or mixed classes.

1.5 Water Potability Dataset [2]

Dataset details: 9 features, 2 classes, 3276 points.

Number of empty sectors in last ring (9-dimensional subspace): 93

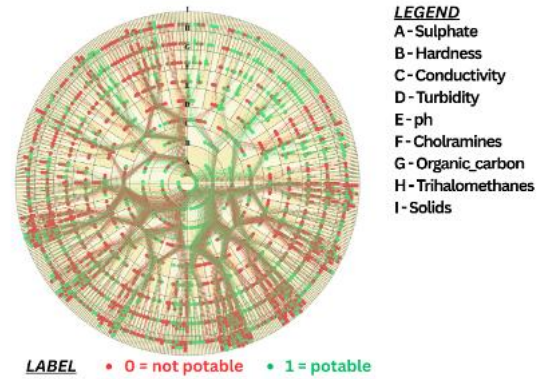


Figure 21: Water Potability Dataset subspaces with empty sectors.

Figure 21 shows the Water Potability Dataset using the RAYS representation, with the full set of occupied and empty sectors visible, representing the spread of data in the R^d space. The density of the points and edges suggests that a larger number of points lie in one half of the space (bottom). However, the distribution of classes across the space seems similar to each other.

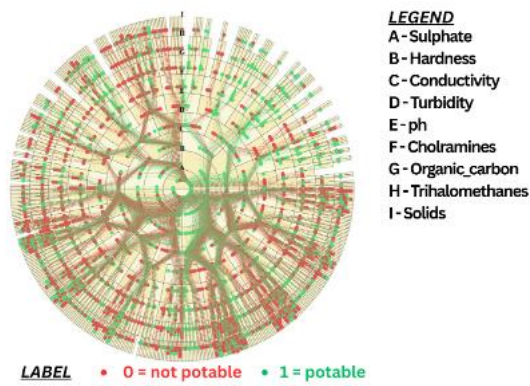


Figure 22: Compact view of the Water Potability Dataset with empty sectors.

Figure 22 offers a condensed view of the same sector structure by removing the empty sectors. This view shows that most of the orthants are occupied in the outermost ring, showing the orthant usage and sparsity of points.

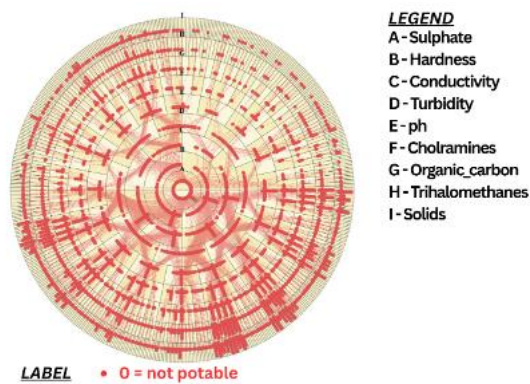


Figure 23: Water Potability Dataset subspaces highlighting label 0.

Figure 23 shows only class label 0 (by using a class filter due to a high number of points), making it easier to analyse how one class is distributed across the sector hierarchy. In this case, observe that a larger number of points lie in the lower half (more density of points in the lower half) as compared to the upper half (less dense). However, the point distribution is more or less evenly spread across the lower half.

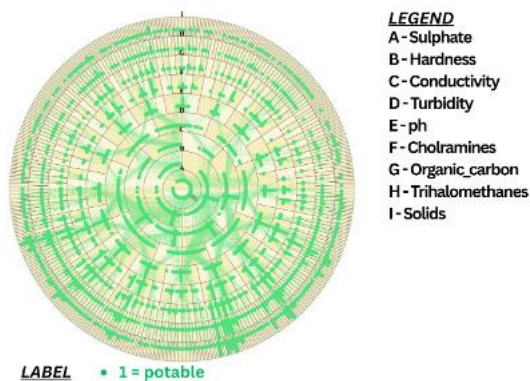


Figure 24: Water Potability Dataset subspaces highlighting label 1.

Figure 24 shows only class label 1 (by using a class filter due to a

high number of points), making it easier to compare how the second class is distributed across the sector hierarchy and to identify dense and sparse regions.

REFERENCES

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